<u>Intradermal coinfection of mice ears with *L. major* IA-2 and *Staphylococcus aureus* Newman strain (MSSA)</u>

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Materials:

- 6-8-week-old female C57Bl/6 mice from Charles River
- *S. aureus* Newman
- *L. major* IA-2
- Sterile tryptic soy broth and tryptic soy agar plates
- Schneider's *Drosophila* medium + *L*-glutamine (Gibco by Life Technologies) supplemented with 10% heat-inactivated fetal bovine serum (FBS) (SAFC Industries), 2 mM *L*-glutamine (Gibco by Life Technologies), 50 μg/mL gentamicin sulfate (IBI Scientific), and 1.2 μg/mL biopterin (Cayman Chemical Company)
- 40% Ficoll-Paque PLUS (GE Healthcare) (40% Ficoll, 50% deionized water, 10% 10X PBS, 0.22 μm filter sterilized, protected from light, stored at 4°C)
- 10% Ficoll-Paque PLUS (GE Healthcare) (10% Ficoll, 20% 5X M199 [5.49 g M199 powder, 0.166 g Na bicarbonate, 100 mL deionized water, 0.22 μm filter sterilized, pH 6.85], 70% deionized water)
- Ketamine (80 mg/kg, Ketalar®, Par Pharmaceutical Cos., Inc.) and xylazine (10 mg/kg, AnaSed®, LLOYD Laboratories, LLOYD Inc.)
- 1.5 mL sterile microcentrifuge tubes
- 27 gauge x ½ inch 1 mL needles for intraperitoneal injection
- Mitutoyo Flat Anvil Dial Thickness Gage (0-22 mm) in 0.01 mm increments
- Caliper and ruler
- 0.37 x 12.7 mm insulin syringe for intradermal injection
- 15 mL or 50 mL conical or 1-inch long piece of plastic tubing
- Phosphate buffered saline (PBS)
- Spectrophotometer
- 37°C incubator and shaking incubator

Methods:

S. aureus preparation:

Day -2: Streak a TSA plate with *S. aureus* Newman MSSA, incubate overnight (O/N) at 37°C.
Day -1: Inoculate an isolated colony from TSA plate in 5 mL of TSB and incubate with shaking at 37°C O/N.
Day 0:

1. Subculture 250 μL of overnight *S. aureus* culture in 25 mL of fresh TSB ~1.5-2 hours to an OD of 0.5 in 37°C shaking incubator.
2. Transfer 10 mL of culture into 50 mL conical tube containing 40 mL PBS. Centrifuge at 3500 rpm for 10 minutes.

3. Resuspend pelleted *S. aureus* cells in sterile PBS to achieve a concentration of $1x10^5$ colony-forming units (CFU)/mL (Note: using our spectrophotometer, OD of 1 = $3x10^8$ CFU/ml = $3x10^5$ CFU/μL).
4. Dilute in PBS to the following concentrations and place on ice:
   - $2x10^3$ CFU/μL (so that 5 μL is 10^4) – to be mixed 1:1 with *L. major* for *L. major* 10^6 parasites + *S. aureus* 10^4 CFUs injection
   - $1x10^3$ CFU/μL (so that 10 μL is 10^4) – for *S. aureus* 10^4 injection
5. For confirmation of infectious dose, plate serial dilutions of inoculum on TSA plates and count ensuing colonies after overnight incubation at 37°C.

***L. major* preparation:**

1. Isolate metacyclic parasites from 30 mL of 3-5 day-old *L. major* IA-2 cultures grown in Schneider's medium by 40% and 10% Ficoll gradient as described previously (Spath GF, Beverley SM. A lipophosphoglycan-independent method for isolation of infective Leishmania metacyclic promastigotes by density gradient centrifugation. Exp Parasitol. 2001;99(2):97-103. doi: 10.1006/expr.2001.4656. PubMed PMID: 11748963.)
2. Wash parasites in PBS and resuspend in PBS at the following dilutions:
   - $2x10^5$ parasites/μl (so that 5 μL contains $1x10^6$ parasites) - to be mixed 1:1 with *S. aureus* for *L. major* 10^6 parasites + *S. aureus* 10^4 CFUs injection
   - $1x10^5$ parasites/μl (so that 10 μL contains $1x10^6$ parasites) - for *L. major* 10^6 injection

Intradermal ear injection:

1. Anesthetize mice with ketamine/xylazine mix in PBS by intraperitoneal injection.
2. Measure thickness of right ear using thickness gage and lesion width and length using caliper and ruler
3. Load insulin syringe with 10 μL of PBS, *L. major*, *S. aureus,* or *L. major* + *S. aureus* mixes.
4. Inject 10 μL intradermally into the right ear pinna of mice using a 15 mL or 50 mL conical tube or a piece of plastic tubing as support. Syringe needles should be bevel up and visible under the skin. Inoculum delivery should result in clear formation of a bubble at the injection site.
5. Mark tail with a permanent marker to identify within group.
6. Record amount of ketamine used and when mice begin and recover from anesthesia.

Infection monitoring:

Measure lesion area by caliper and thickness days 1-7, 14, 21, and 28, and weekly thereafter if desired.

Euthanasia:

1. At experiment endpoint, euthanize mice using CO_2 asphyxiation followed by cervical dislocation in accordance with University of Iowa IACUC guidelines.
2. Harvest and process injected ears, draining cervical/submandibular lymph nodes, and/or other organs for desired experimental studies.
3. Package euthanized animals for incineration protocol.